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Permutation P -values Should Never Be Zero: Calculating Exact P -values When Permutations Are Randomly Drawn

Belinda Phipson and Gordon K. Smyth

Abstract

Permutation tests are amongst the most commonly used statistical tools in modern genomic research, a process by which p -values are attached to a test statistic by randomly permuting the sample or gene labels. Yet permutation p -values published in the genomic literature are often computed incorrectly, understated by about $1/m$, where m is the number of permutations. The same is often true in the more general situation when Monte Carlo simulation is used to assign p -values. Although the p -value understatement is usually small in absolute terms, the implications can be serious in a multiple testing context. The understatement arises from the intuitive but mistaken idea of using permutation to estimate the tail probability of the test statistic. We argue instead that permutation should be viewed as generating an exact discrete null distribution. The relevant literature, some of which is likely to have been relatively inaccessible to the genomic community, is reviewed and summarized. A computation strategy is developed for exact p -values when permutations are randomly drawn. The strategy is valid for any number of permutations and samples. Some simple recommendations are made for the implementation of permutation tests in practice.

KEYWORDS: permutation test, Monte Carlo test, p -values, multiple testing, microarray

Erratum

The following information was added February 17, 2011:

The published article claimed that using an estimated $\hat{p}$ -value, in place of the actual p -value, increases the type I error rate. While this is typically true, there are special combinations of m and α , for which the obvious unbiased estimator $\hat{p}$ for the p -value does result in a Monte Carlo test of the correct size. The following corrections are intended to clarify this. The recommendations of the article are unchanged.

Page 1, last line, “ p -values increases” should read “ p -values often increases”.

Page 2, Section 2 title, “Estimating p -values increases the type I error rate” should read “Estimating p -values tends to increase the type I error rate”.

Page 3, third line, the word “will” should read “can”.

Page 3, line 9, the word “generally” should read “often”.

Page 3, fourth line from the bottom, the two sentences “This always exceeds the nominal size α . The type I error rate in fact never falls below $1/(m+1)$ regardless of how stringent the desired rate α is chosen.” should read:

“This expression can be greater or less than the nominal size α . The discreteness of $\hat{p}$ however ensures that $P(\hat{p} \leq \alpha) > \alpha$ for most values of α near zero, and $P(\hat{p} \leq \alpha) < \alpha$ for most values of α near 1 (Figure A). Since $\alpha \leq 0.05$ for most experiments, the true type I error rate $P(\hat{p} \leq \alpha)$ exceeds α for most practical values of m and α . The type I error rate always exceeds α when $\alpha = i/m$ for some integer i , or when $\alpha < 1/(m+1)$. In fact $P(\hat{p} \leq \alpha)$ is never less than $1/(m+1)$ regardless of how stringent the desired rate α is chosen.”

Additional Figure A is given below.

Page 3, last line, “does not lead to an test” should read “does not generally lead to a test”.

Page 4, first line, the word “systematically” should read “usually”.

Page 10, Section 7, second paragraph, the phrase “should never be used, because they fail to control the type I error rate” should read “should not be used, because they can’t be guaranteed to control the type I error rate”.

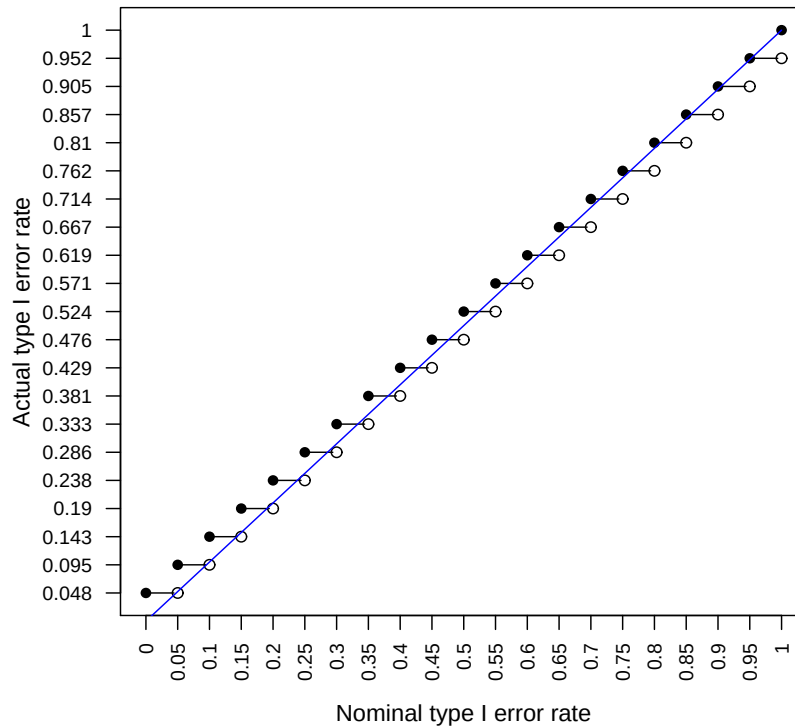


Figure A: Type I error rate, $P(\hat{p} \leq \alpha)$, arising from using $\hat{p}$ as a working p -value (y-axis) versus the nominal size α (x-axis) when $m = 20$. The equality line, $P(\hat{p} \leq \alpha) = \alpha$, is shown in blue. The type I error rate is generally above the nominal rate for $\alpha < 0.5$.

1 Introduction

A permutation test is a type of non-parametric randomization test in which the null distribution of a test statistic is estimated by randomly permuting the class labels of the observations. Permutation tests are highly attractive because they make no assumptions other than that the observations are independent and identically distributed under the null hypothesis. They are often used when standard distributional assumptions are doubtful or when the null distribution of the test statistic is simply intractable. This article also considers Monte Carlo tests, by which we mean simulation procedures in which pseudo-random instances of a test statistic are generated from its null distribution.

The idea of permutation tests was first described by Fisher (1935), with the assumption that all possible permutations would be enumerated. The idea of limiting the computations to a subset of permutations, randomly drawn from the set of possible permutations, was first examined by Dwass (1957). The idea of Monte Carlo tests seems to have been first suggested by Barnard (1963). Each of these three original papers included an expression for the exact p -value arising from the randomization test. None of the expressions were suitable for practical computation at the time, however, and the results on randomly drawn permutations seem to have been largely forgotten by the literature. Dwass (1957) in particular is not easily readable by applied statisticians, and we have not seen the exact p -value expression from Dwass (1957) repeated in any subsequent paper. Even recent reviews that cite Dwass (1957) do not reproduce the exact p -value (Ernst, 2004).

Recent decades have seen an enormous development of bootstrap and re-sampling inference techniques (Efron, 1982; Efron and Tibshirani, 1993; Davison and Hinkley, 1997; Manly, 1997). A key concept of this research is the idea that the distribution of an estimator or test statistic can be estimated by an empirical re-sampling distribution. While this research has been tremendously valuable for a multitude of purposes, we believe it has had an unfortunate influence on the implementation of permutation tests. Rather than using the exact p -value expressions from the original papers mentioned above, it has become common practice to use estimators of p -values in place of genuine probabilities. This estimation approach is recommended in many textbooks (Higgins, 2004; Efron and Tibshirani, 1993; Good, 1994; Mielke and Berry, 2001). Some recent papers have given confidence intervals for permutation p -values (Ernst, 2004), again re-inforcing the idea of p -value calculation as an estimation problem. The view is often expressed that the correct p -values can be estimated to any required precision by drawing a sufficiently large number of permutations.

This article points out this is only true up to a point. We show that substituting unbiased estimators for exact p -values increases the type I error rate of

the test. A similar concern was raised by Onghena and May (1995) and Edgington and Onghena (2007). We also show that, when a very large number of simultaneous tests is to be done, p -values can never be estimated with sufficient precision to avoid serious type I error problems at the family-wise level. Meanwhile, exact p -values that do control the type I error rate correctly are readily available for little or no additional computational cost.

With the development of large genomic data sets and modern computing facilities, permutation tests have become one of the most commonly used statistical tools in modern biostatistical and genomic research. A google scholar search for “gene” and “permutation test” finds nearly 8000 genomic articles, likely an underestimate of the number of relevant articles. Permutations are almost always randomly drawn rather than exhaustively enumerated. We were motivated to write this article because of the number of times we have seen permutation p -values computed incorrectly in the genomic literature. Many published algorithms and software tools return permutation “ p -values” that are really estimates of p -values, i.e., $\hat{p}$ -values. In most cases, published p -values obtained from permutation tests are understated by about $1/m$, where m is the number of permutations. The situation is similar when Monte Carlo simulation is used to assign p -values. Of particular concern is the tendency of many public software packages to return $\hat{p}$ -values that can be exactly zero. It is obvious that a permutation test could never yield an exactly zero p -value if all permutations were exhaustively enumerated, and it makes no inferential sense to assert that the p -value can be reduced to zero by considering only a subset of the permutations. Most concerning of all is the potential for multiple testing procedures to be applied to $\hat{p}$ -values that are exactly zero. This has the potential to lead to grossly misleading results at the family-wise level, in which the results are judged to be significant at fantastically strong levels, even when all of the null hypotheses are true.

This article contains only a modest amount of novel statistical theory. We provide a new computational solution for evaluating exact p -values when permutations are drawn at random. Our main aim is to clarify the need to evaluate exact p -values, and to make the relevant results more accessible to the genomic community.

2 Estimating p -values increases the type I error rate

Randomization tests are used in the following context. A test statistic t has been calculated from the data at hand. We would like to use the null hypothesis tail probability $p_\infty = P(T > t)$ of the observed test statistic as the p -value, but this probability cannot be directly computed. In a general sense, the purpose of randomization tests

is to estimate p_∞ . The purpose of this section is to point out that p -value calculation must take into account not just the estimate of p_∞ but also the margin of error of this estimate. Simply replacing p_∞ with an unbiased estimator will result in a failure to correctly control the type I error rate.

If the test statistic t is continuously distributed, then p_∞ will be uniformly distributed on $(0, 1)$ under the null hypothesis. To control the type I error rate at level α , if p_∞ was known, the null hypothesis would be rejected when $p_\infty \leq \alpha$. In other words, $P(p_\infty \leq \alpha) = \alpha$ under the null hypothesis.

Now suppose that $\hat{p}$ is an unbiased estimator of p_∞ . It is generally true that $P(\hat{p} \leq \alpha) > \alpha$, because $\hat{p} < \alpha$ occurs with positive probability even when $p_\infty > \alpha$. In general

$$P(\hat{p} \leq \alpha) = \int_0^1 P(\hat{p} \leq \alpha | p_\infty) dp_\infty.$$

To consider a concrete case, suppose that m Monte Carlo data sets are generated under the null hypothesis, and suppose that $\hat{p} = B/m$ where B is the number of Monte Carlo test statistics greater than or equal to t . Then $m\hat{p}$ is binomially distributed with size m and success probability p_∞ . Plainly, $\hat{p}$ is an unbiased estimator of p_∞ . Furthermore, $\hat{p}$ converges to p_∞ as the number of Monte Carlo data sets $m \rightarrow \infty$. This is all conditional on the observed data however. To compute the size of the resulting Monte Carlo hypothesis test, we have to average over all possible data sets that might have been observed. Since $m\hat{p}|p_\infty$ is binomial and p_∞ is uniform, the unconditional distribution of $m\hat{p}$ is a special case of beta-binomial. In fact, it is easy to show that $\hat{p}$ follows a discrete uniform distribution on the values $0, 1/m, 2/m, \dots, 1$:

$$\begin{aligned} P(\hat{p} = b/m) &= \int_0^1 P(\hat{p} = b/m | p_\infty) dp_\infty \\ &= \int_0^1 \binom{m}{b} p_\infty^b (1 - p_\infty)^{m-b} dp_\infty \\ &= \frac{1}{m+1} \end{aligned}$$

for all $b = 0, 1, \dots, m$. Hence the type I error rate of the Monte Carlo test that arises from using $\hat{p}$ as a working p -value is

$$P(\hat{p} \leq \alpha) = \frac{\lfloor m\alpha \rfloor + 1}{m+1}.$$

This always exceeds the nominal size α . The type I error rate in fact never falls below $1/(m+1)$ regardless of how stringent the desired rate α is chosen.

This demonstrates an unintuitive principle. Inserting an unbiased estimator of a p -value for the true p -value does not lead to a test of the same size. Rather,

the type I error rate of the resulting randomized test is systematically greater than that which would arise from using the true p -value. To achieve a randomized test of the correct size, we must in effect substitute for the true p -value an estimator with positive bias. The moral is that p -values must be directly evaluated as probabilities, not treated as if they were unknown parameters to be estimated in a statistical model.

3 Under-estimating p -values is dangerous in a multiple testing context

If only one hypothesis is to be tested, mis-calculating a p -value probability by a small amount will often be of little consequence. In genomic research, however, it is typically the case that many tests are to be conducted. When the number of tests is large, any systematic under-estimation of p -values can lead to dangerously wrong conclusions at the family-wise level. Suppose for example that 50,000 tests are to be conducted, one for each transcript in a transcriptome perhaps, and suppose that the resulting p -values are to be adjusted by Bonferroni or Holm's methods (Holm, 1979). If each p -value is under-estimated by just $1/50,000$, then it is highly likely that at least one null hypothesis will be rejected at any reasonable significance level, even if none of the null hypotheses are false. In this case the family-wise type I error rate is radically wrong, even though the type I error rate is only very slightly under-estimated for each individual test.

This situation can arise easily if a permutation test is conducted for each gene on a microarray platform. To give an example, suppose that permutation $\hat{p}$ -values are computed from $m = 1000$ sample permutations for each of 30,000 genes, where the total number of distinct permutations possible is much larger than 1000. Suppose that a working p -value is computed for each gene as $\hat{p} = B/m$, where B is the number of permutations yielding a test statistic at least as extreme as that observed from the original data. Clearly $\hat{p} = B/m$ is an unbiased estimate of the tail probability for the test statistic under the null distribution induced by permutating samples. Yet then the number of exact zero values will be around 30, even when there are no differentially expressed genes. These 30 zero values will remain statistically significant regardless of multiple testing procedure or significance level applied.

The basic problem here is that the family-wise error rate depends on the relative accuracy on the p -values, rather than on additive errors. For the smallest p -values, the error may be large in relative terms even when it is small in absolute terms. When a non-zero p -value is mis-estimated as zero, the relative error is 100%.

4 Exact p -values for Monte Carlo tests

It is helpful to consider first the case of Monte Carlo tests, which have much in common with permutations tests, but are mathematically simpler. Suppose t_{obs} is the observed test statistic for a given hypothesis for a given experiment. The basic assumption of Monte Carlo tests is that it is possible to generate independent random datasets under the null hypothesis H_0 . Typically, this is done by simulation using pseudo-random numbers. Suppose m such datasets are simulated under the null hypothesis, each yielding a distinct test statistic t_{sim} . In principle, the ideal p -value is

$$p_{\infty} = P(t_{\text{sim}} \geq t_{\text{obs}})$$

but this is unknown. To compute p_{∞} exactly it would be necessary to generate an infinite number of simulated datasets, but only a finite number m are available.

For simplicity, we assume that the test statistic t is continuous, so that p_{∞} is uniform on the unit interval and all the t_{sim} are distinct. Let B be the number of times out of m that $t_{\text{sim}} \geq t_{\text{obs}}$. The previous section showed that using the unbiased estimator $\hat{p}_{\infty} = B/m$ in place of p_{∞} leads to an invalid test that does not correctly control the type I error rate at the required level. A more valid approach is to compute the tail probability directly for the Monte Carlo results. From the point of view of the randomization test, the test statistic is B rather than t_{obs} , and the required tail probability is $P(B \leq b)$. The previous section showed that, under the null hypothesis, the marginal distribution of B over all possible data sets is discrete uniform on the integers from $0, \dots, m$. Hence the exact Monte Carlo p -value is

$$p_u = P(B \leq b) = \frac{b+1}{m+1}.$$

In effect, the randomization test replaces the unknown tail probability p_{∞} with a biased estimator $(b+1)/(m+1)$, the amount of positive bias being just enough to allow for the uncertainty of estimation and to produce a test with the correct size. This p -value calculation is presented in a number of textbooks on randomization tests including Davison and Hinkley (1997), Manly (1997) and Edgington and Onghena (2007).

5 Sampling permutations without replacement

Now we consider permutation tests, a type of re-sampling randomization test. The basic setup for a permutation test is as follows. We observe data pairs (y_i, g_i) , $i = 1, \dots, n$, where the y_i are independent data observations and the g_i are labels

that classify the observations into two or more groups. The aim of a permutation test is to test the association between y and g .

A suitable test statistic t is computed from the (y_i, g_i) . This might be a two-sample t -statistic, for example, but it could in principle be any statistic designed to test for differences between the data groups. Write t_{obs} for the observed value of the statistic. The idea of a permutation test is to assign a p -value to t_{obs} by randomly permuting the class labels g_i a large number of times. Then the empirical distribution of the test statistics t_{perm} from the permuted data estimates the null distribution of the test statistic. The null hypothesis H_0 being tested is that the y_i are independently and identically distributed, versus an alternative hypothesis in which the distribution of y_i differs between groups.

For simplicity, we will assume that the y_i are all distinct, so that permutation yields distinct values of the test statistic unless the partition of y values into groups is identical. We assume that there are m_t possible distinct values of the test statistic, not counting the original value, and that all are equally likely to arise under the permutation distribution.

The original idea of permutation tests was that all possible permutations would be enumerated (Fisher, 1935). However m_t can be very large indeed for moderate to large sample sizes, so that complete enumeration may be impractical. In this situation, it is common practice to examine a random subset of the possible permutations. It is important to distinguish two possible scenarios. The random permutations may be drawn without replacement, so that all permutations are distinct. Or permutations may be drawn with replacement, so that the same permutation may be used more than once. We consider the first scenario now. The second scenario is the topic of the next section.

Suppose that a random sample $m \leq m_t$ permutations is drawn, without repetition, such that each permutation yields a distinct test statistic t_{perm} , different from the original statistic. Then the situation is essentially the same as for Monte Carlo tests considered in the previous section. The argument used in the previous section to derive the Monte Carlo p -value applies again, showing that the exact permutation p -value is $p_u = (b + 1)/(m + 1)$, where b is the number of t_{perm} greater than t_{obs} .

This result includes the situation in which all m_t possible distinct permutations are evaluated. The exhaustive permutation p -value is $p_t = (b_t + 1)/(m_t + 1)$, where b_t is the total number of distinct statistic values exceeding t_{obs} .

6 Sampling permutations with replacement

6.1 Exact p -values

We now consider the case in which permutations are randomly drawn with replacement. Each permutation is randomly generated independently of previous permutations or of the original data configuration. This is by far the most common way that permutation tests are used in practice, especially in the genomic literature. This scenario requires careful treatment from a mathematical point of view, because of the possibility that the random permutations may include repetitions of the same permutations and indeed of the original data.

Suppose that an independent random sample of m permutations is drawn with replacement. The resulting test statistics t_{perm} may include repeat values, including possibly the original observed value t_{obs} . The exact p -value is now slightly less than $(b+1)/(m+1)$, because of the possibility that the original data is included at least once as one of the random permutations.

Write B for the number of permutations out of m yielding test statistics at least as extreme as t_{obs} . We wish to evaluate the exact permutation p -value $p_e = P(B \leq b)$ for any given b . We do so using a formal conditional approach.

Write B_t for the total number of possible distinct statistic values exceeding t_{obs} , again an unknown quantity, and note $p_t = (B_t + 1)/(m_t + 1)$. Although p_t is in principle the ideal permutation p -value, it is unknown in the scenario we now consider. If the null hypothesis is true, then B_t follows a discrete uniform distribution on the integers $0, \dots, m_t$. Conditional on $B_t = b_t$, B follows a binomial distribution with size m and probability p_t . Hence

$$p_e = \sum_{b_t=0}^{m_t} P(B \leq b | B_t = b_t) P(B_t = b_t | H_0) = \frac{1}{m_t + 1} \sum_{b_t=0}^{m_t} F(b; m, p_t) \quad (1)$$

where F is the cumulative probability function of the binomial distribution. This expression is essentially the same as Proposition 1 of Dwass (1957), although Dwass did not seem to intend it for practical computational use. Although not very difficult to derive, this exact p -value has not been repeated to our knowledge elsewhere in the literature.

6.2 Computation and approximation

Equation (1) provides a formula for the exact p -value, but the summation may not be practical to evaluate explicitly if m_t is very large. Hence we explore an approx-

imation that can be evaluated quickly. Replacing the summation with an integral, and using a continuity correction, yields

$$\begin{aligned} p_e &\approx \int_{0.5/(m_t+1)}^1 F(b; m, p_t) dp_t \\ &= \int_0^1 F(b; m, p_t) dp_t - \int_0^{0.5/(m_t+1)} F(b; m, p_t) dp_t \\ &= \frac{b+1}{m+1} - \int_0^{0.5/(m_t+1)} F(b; m, p_t) dp_t. \end{aligned} \quad (2)$$

This shows that $p_e < p_u = (b+1)/(m+1)$, and also that p_e converges to p_u as m_t increases. The last integral in equation (2) is easily evaluated using numerical integration algorithms. We have found it convenient to use Gaussian quadrature.

The valid but conservative p -value p_u has been recommended by a number of authors including Davison and Hinkley (1997), Manly (1997) and Ernst (2004). The extent to which p_u over-estimates the exact p -value depends on m_t . When $m_t = 1000$, p_u over-estimates the 0.05 p -value by about 2%, but smaller p -values are over-estimated by greater relative amounts (Figure 1b). When m_t is small, p_u can be very conservative (Figure 1a).

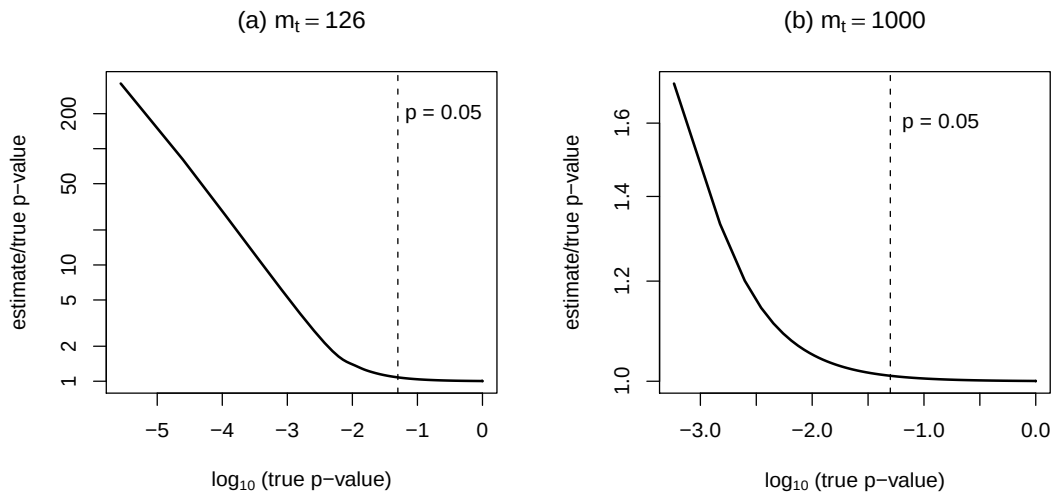


Figure 1: Ratio of the upper bound $p_u = (b+1)/(m+1)$ to the exact p -value. Panel (a) is for $m_t = 126$ and (b) is for $m_t = 1000$. The number of permutations is $m = 1000$ in each case, and values are plotted for all possible values of b .

Table 1: Exact and approximate p -values for a one-sided test between two groups of size five using $m = 100$ random permutations.

Number of $t_{\text{sim}} \geq t_{\text{obs}}$	Estimated $\hat{p}$ -value	Exact p -value	Integral approximated p -value	Upper bound p_u
0	0	0.008047755	0.008101416	0.0099
1	0.01	0.017818517	0.017829558	0.0198
2	0.02	0.027718516	0.027719402	0.0297
3	0.03	0.037619829	0.037619855	0.0396
4	0.04	0.047520825	0.047520824	0.0495
5	0.05	0.057421814	0.057421814	0.0594
6	0.06	0.067322804	0.067322804	0.0693
7	0.07	0.077223794	0.077223794	0.0792
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

6.3 Specific scenarios

The number m_t of possible distinct values of the test statistic is not simply a function of the number of permutations, but also depends on the form of the test statistic and on the sample sizes. This is because different permutations can lead to the same test statistic, especially for two-sided tests with balanced samples.

Consider a comparison between two groups of sizes n_1 and n_2 . In this situation it is usual to sample permutations by randomly assigning n_1 of the indices i to group 1 and the remaining n_2 to group 2. If the test statistic is one-sided, or if $n_1 \neq n_2$, then each permutation will generally yield a distinct test statistic. In that case, the number of distinct values of the permuted statistics is

$$m_t + 1 = \binom{n_1 + n_2}{n_1}.$$

For example, a one-sided test with $n_1 = n_2 = 5$ gives $m_t + 1 = \binom{10}{5} = 252$. The integral approximation (2) provides an excellent approximation to the exact p -values (Table 1).

If a two-sided test statistic is used, for example the absolute value of a t -statistic, and if the two groups are of equal size $n_1 = n_2$, then each permutation gives the same value of the test statistic as the corresponding permutation with the two groups reversed. In this case, the total number of distinct test statistics possible is $m_t + 1 = \binom{n_1 + n_2}{n_1} / 2$.

If there are three or more groups of unequal sizes then $m_t + 1$ is the multinomial coefficient

$$m_t + 1 = \binom{\sum_{j=1}^k n_j}{n_1 \dots n_k}$$

where k is the number of groups. If some of the group sizes n_j are equal, then some permutations will in general give equal test statistics, so the calculation of m_t will need to consider the particular configuration of samples sizes.

7 Discussion and recommendations

The approach posited in Section 5, of drawing permutations without repeating previous values of the test statistic, turns out to be a surprisingly difficult combinatorial programming exercise. Hence this approach is seldom used—so much so that we have never seen it in the genomic literature. Nevertheless, this is the superior approach to permutation tests. Sampling without replacement gives rise to a simple easily understood exact p -value in the form of p_u . Even more importantly, it is possible to show that sampling permutations without replacement gives more statistical power than sampling with replacement for any given number of permutations $m \leq m_t$. Our first recommendation then is that more attention be given to this problem by computer scientists and combinatorialists, to provide biostatisticians with efficient numerical algorithms to sample permutations without replacement.

Our second recommendation is that “unbiased” estimators of p -values such as $\hat{p}$ should never be used, because they fail to control the type I error rate. Although the risk involving in using $\hat{p}$ in place of the exact p -value may be modest in many applications, there is no justification for using an invalid p -value when a valid p -value, in the form of p_u , is readily available for no extra cost.

Our third recommendation is that users might consider using the exact p -value p_e , at the cost of a trivial amount of extra computation. Our algorithm to compute p_e is implemented as the function `permp` in the software package `statmod` for R available from <http://www.r-project.org>. The exact value p_e gives a worthwhile gain in statistical power over p_u when the number of permutations m performed is a more than a negligible proportion of the total number possible m_t .

A referee asked us to warn users that the number of permutations m should not be chosen too small. While this is true, the urgent need to avoid small m disappears when the exact p -value p_e is used in place of $\hat{p}$, because p_e ensures a valid statistical test regardless of sample size or the number of permutations. When exact p -values are used, the only penalty for choosing m small is a loss of statistical power to reject the null hypothesis. This is as it should be: more permutations should generally provide greater power.

The same referee asked us to comment on the difference between permuting genes and permuting samples. One obvious difference between genes and samples is that there are generally more of the former than the latter, so that m_t is likely to be astronomically large when genes are permuted. This implies that p_u is likely to be a good approximation for p_e in this context. We have avoided discussing gene permutation in this article however because gene permutation has another obvious problem. Permutation assumes that the items being permuted are statistically independent, whereas genes usually cannot be assumed independent in genomic applications (Goeman and Bühlman, 2007; Wu et al, 2010). In general, we recommend permuting samples rather than permuting genes when there is a choice.

Finally, it is worthwhile mentioning that there are alternatives to permutation that alleviate the problem of small m_t when the number of samples is not large (Dørum, 2009; Wu et al, 2010).

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